$$\underline{\sigma}_{n+1} = \underline{\sigma}^{vol}_{n+1} + \underline{\sigma}^{iso}_{n+1}$$

$$\Delta\underline{\sigma}_{n+1} = \Delta\underline{\sigma}^{vol}_{n+1} + \Delta\underline{\sigma}^{iso}_{n+1}$$

$$\underline{\sigma}^{vol}_{n+1} = \underline{\mathbb{C}}^{e,vol} : \underline{\varepsilon}^{e}_{n+1} \longrightarrow (1)$$

$$\underline{\sigma}^{iso}_{n+1} = \underline{\mathbb{C}}^{e,iso} : \underline{\varepsilon}^{e}_{n+1} \longrightarrow (2)$$

$$\Delta\underline{\sigma}^{vol}_{n+1} = \frac{\partial \underline{\sigma}^{vol}_{n+1}}{\partial \underline{\varepsilon}^{e}_{n+1}} : \frac{\partial \underline{\varepsilon}^{e}_{n+1}}{\partial \underline{\varepsilon}_{n+1}} : \Delta\underline{\varepsilon} \longrightarrow (3)$$

($\frac{\partial \underline{\sigma}^{vol}_{n+1}}{\partial \underline{\varepsilon}^{e}_{n+1}} = \underline{\mathbb{C}}^{e,vol}$)

$$\Delta\underline{\sigma}^{iso}_{n+1} = \frac{\partial \underline{\sigma}^{iso}_{n+1}}{\partial \underline{\varepsilon}^{e}_{n+1}} : \frac{\partial \underline{\varepsilon}^{e}_{n+1}}{\partial \underline{\varepsilon}_{n+1}} : \Delta\underline{\varepsilon} \longrightarrow (4)$$

($\frac{\partial \underline{\sigma}^{iso}_{n+1}}{\partial \underline{\varepsilon}^{e}_{n+1}} = \underline{\mathbb{C}}^{e,iso}$)

$$\frac{\partial \underline{\varepsilon}^{e}_{n+1}}{\partial \underline{\varepsilon}_{n+1}} = ?$$

$$\underline{\varepsilon}^{e}_{n+1} = \underline{\varepsilon}_{n+1} - \underline{\varepsilon}^{p}_{n+1}$$

$$\underline{\varepsilon}^{e}_{n+1} = \underline{\varepsilon}_{n+1} - \underline{\varepsilon}^{p}_{n} - \Delta\lambda\, \underline{n}^{trial}_{n+1}$$

$$\underline{\varepsilon}^{e}_{n+1} = \underline{\varepsilon}^{e,trial}_{n+1} - \Delta\lambda\, \underline{n}^{trial}_{n+1}$$

$$\underline{\varepsilon}^{e}_{n+1} = \underline{\varepsilon}^{e,trial}_{n+1} - \frac{\phi^{trial}}{\left(\frac{2}{3}H + 2\mu\right)}\, \underline{n}^{trial}_{n+1} \longrightarrow (5)$$

$$\underline{\varepsilon}^{e}_{n+1} = \underline{\varepsilon}^{e}_{n+1}\left(\underline{\varepsilon}^{e,trial}_{n+1},\ \phi^{trial},\ \underline{n}^{trial}_{n+1}\right)$$

$$\frac{\partial \underline{\varepsilon}^{e}_{n+1}}{\partial \underline{\varepsilon}_{n+1}} = \underbrace{\frac{\partial \underline{\varepsilon}^{e}_{n+1}}{\partial \underline{\varepsilon}^{e,trial}_{n+1}}}_{a} : \underbrace{\frac{\partial \underline{\varepsilon}^{e,trial}_{n+1}}{\partial \underline{\varepsilon}_{n+1}}}_{b} + \underbrace{\frac{\partial \underline{\varepsilon}^{e}_{n+1}}{\partial \phi^{trial}}}_{c} \otimes \underbrace{\frac{\partial \phi^{trial}}{\partial \underline{\varepsilon}_{n+1}}}_{d} + \underbrace{\frac{\partial \underline{\varepsilon}^{e}_{n+1}}{\partial \underline{n}^{trial}_{n+1}}}_{e} : \underbrace{\frac{\partial \underline{n}^{trial}_{n+1}}{\partial \underline{\varepsilon}_{n+1}}}_{f} \longrightarrow (6)$$

(a) $$\frac{\partial \underline{\varepsilon}^{e}_{n+1}}{\partial \underline{\varepsilon}^{e,trial}_{n+1}} = \underline{\underline{\mathbb{I}}}$$

(b) $$\underline{\varepsilon}^{e,trial}_{n+1} = \underline{\varepsilon}_{n+1} - \underline{\varepsilon}^{p}_{n}$$

$$\frac{\partial \underline{\varepsilon}^{e,trial}_{n+1}}{\partial \underline{\varepsilon}_{n+1}} = \underline{\underline{\mathbb{I}}}$$

(c) $$\frac{\partial \underline{\varepsilon}^{e}_{n+1}}{\partial \phi^{trial}} = -\frac{\underline{n}^{trial}_{n+1}}{\left(\frac{2H}{3} + 2\mu\right)}$$

(d) $$\phi^{trial} = \| \mathrm{dev}\, \underline{\sigma}^{trial}_{n+1} \| - \sqrt{\frac{2}{3}} \left(y_0 + H\alpha_n\right)$$

$$\phi^{trial} = 2\mu \| \mathrm{dev}\, \underline{\varepsilon}^{e,trial}_{n+1} \| - \sqrt{\frac{2}{3}} \left(y_0 + H\alpha_n\right)$$

$$\frac{\partial \phi^{trial}}{\partial \underline{\varepsilon}^{e,trial}_{n+1}} = \frac{2\mu\, \mathrm{dev}\, \underline{\varepsilon}^{e,trial}_{n+1}}{\| \mathrm{dev}\, \underline{\varepsilon}^{e,trial}_{n+1} \|} : \underline{\underline{\mathbb{P}}}$$

$$\frac{\partial \phi^{trial}}{\partial \underline{\varepsilon}^{e,trial}_{n+1}} = 2\mu\, \underline{n}^{trial}_{n+1}$$

(e) $$\frac{\partial \underline{\varepsilon}^{e}_{n+1}}{\partial \underline{\sigma}^{trial}_{n+1}} = \frac{-\phi^{trial}}{\left(\frac{2H}{3} + 2\mu\right)} \underline{\underline{\mathbb{I}}}$$

(f) $$\underline{n}^{trial}_{n+1} = \frac{\mathrm{dev}\, \underline{\varepsilon}^{e,trial}_{n+1}}{\| \mathrm{dev}\, \underline{\varepsilon}^{e,trial}_{n+1} \|}$$

$$\frac{\partial \underline{n}^{trial}_{n+1}}{\partial \underline{\varepsilon}^{e,trial}_{n+1}} = \frac{\underline{\underline{\mathbb{P}}}}{\| \mathrm{dev}\, \underline{\varepsilon}^{e,trial}_{n+1} \|} - \frac{\mathrm{dev}\, \underline{\varepsilon}^{e,trial}_{n+1} \otimes \mathrm{dev}\, \underline{\varepsilon}^{e,trial}_{n+1} : \underline{\underline{\mathbb{P}}}}{\| \mathrm{dev}\, \underline{\varepsilon}^{e,trial}_{n+1} \|^3}$$

$$\frac{\partial \underline{n}^{trial}_{n+1}}{\partial \underline{\varepsilon}^{e,trial}_{n+1}} = \frac{\underline{\underline{\mathbb{P}}}}{\| \mathrm{dev}\, \underline{\varepsilon}^{e,trial}_{n+1} \|} : \left[\underline{\underline{\mathbb{I}}} - \underline{n}^{trial}_{n+1} \otimes \underline{n}^{trial}_{n+1} \right]$$

Inserting into (6):

$$\frac{\partial \underline{\varepsilon}^{e}_{n+1}}{\partial \underline{\varepsilon}_{n+1}} = \underline{\underline{\mathbb{I}}} - \frac{\mu}{\frac{H}{3}+\mu}\, \underline{n}^{trial}_{n+1} \otimes \underline{n}^{trial}_{n+1} - \frac{\phi^{trial}}{\frac{2H}{3}+2\mu}\, \frac{\underline{\underline{\mathbb{P}}}}{\| dev\, \underline{\varepsilon}^{trial}_{n+1} \|} : \left[\underline{\underline{\mathbb{I}}} - \underline{n}^{trial}_{n+1} \otimes \underline{n}^{trial}_{n+1} \right] \rightarrow (7)$$

$$\Delta \underline{\sigma}^{vol}_{n+1} = \underline{\underline{\mathbb{C}}}^{e,vol} : \left[\frac{\partial \underline{\varepsilon}^{e}_{n+1}}{\partial \underline{\varepsilon}_{n+1}} \right] : \Delta \underline{\varepsilon} = \underline{\underline{\mathbb{C}}}^{e,vol} : \Delta \underline{\varepsilon} \left(\begin{array}{l} \text{All deviatoric terms} \\ \text{go to zero } (tr(dev)=0 \end{array} \right)$$

$$\Delta \underline{\sigma}^{iso}_{n+1} = \left[2\mu \underline{\underline{\mathbb{P}}} \left[1 - \frac{\phi^{trial}}{\| dev\, \underline{\varepsilon}^{e,trial}_{n+1} \| \left(\frac{2H}{3} + 2\mu \right)} \right] - 2\mu\, \underline{n}^{trial}_{n+1} \otimes \underline{n}^{trial}_{n+1} \left[\frac{\mu}{\mu + \frac{H}{3}} + \frac{\phi^{trial}}{\| dev\, \underline{\varepsilon}^{e,trial}_{n+1} \| \left(\frac{2H}{3} + 2\mu \right)} \right] \right] : \Delta \underline{\varepsilon}$$

$$\underline{\underline{\mathbb{C}}}^{ep,iso}$$

$$\underline{\underline{\mathbb{C}}}^{ep} = \underline{\underline{\mathbb{C}}}^{e,vol} + \underline{\underline{\mathbb{C}}}^{ep,iso}$$